

As we acquire more knowledge, things do not become more comprehensible, but more complex and mysterious.

Write a response in which you discuss the extent to which you agree or disagree with the statement and explain your reasoning for the position you take. In developing and supporting your position, you should consider ways in which the statement might or might not hold true and explain how these considerations shape your position.

Select one entry for the blank. Fill the blank in the way that best completes the text.

_____ is valuable in science; even when a scientific idea is true, it can be misused through grandiosity.

Humility
Experimentation
Patience
Cooperation
Exposure

Select one answer choice.

For each blank select one entry from the corresponding column of choices. Fill all blanks in the way that best completes the text.

Though extremely (i)_____ about her own plans, she allowed her associates no such (ii)_____ and was constantly badgering them for information about what they intended to do next.

Blank (i)

candid
demanding
reticent

Blank (ii)

discretion
continuity
solitude

Select one entry from each column.

For each blank select one entry from the corresponding column of choices. Fill all blanks in the way that best completes the text.

The type of cultural snobbery long associated with painting in the art world, whereby beloved popular works are (i) _____ by critics, has appeared in the popular music industry. Bands with (ii) _____ are critically (iii) _____, whereas the more rarefied and minimally marketed performers can do no wrong in the opinion of the critics.

Blank (i)

disdained
analyzed
contested

Blank (ii)

great ambitions
widespread appeal
eclectic influences

Blank (iii)

ignored
aggrandized
reviled

Select one entry from each column.

Questions 4 and 5 are based on this passage.

Just as in many other societies, self-help texts dealing with relationships are widely sold in Mexican cities. Their widespread circulation indicates Mexicans' desire for guidance on matters of family life and other relationships in a cultural environment characterized by the erosion of traditional patriarchal cultural models. On the one hand, the widely varying accounts of and prescriptions for family life provided by these texts may be understood as reflections of the increasingly diverse and often conflicting cultural models of family life in urban Mexico. On the other hand, self-help texts may also play an important role in shaping these cultural models, insofar as they provide a substantial number of contemporary urban Mexicans with cultural options for making sense of their closest relationships.

The primary purpose of the passage is to

- ☐ offer an alternative to prevailing interpretations of the role of self-help texts in the lives of urban Mexicans
- ☐ reflect on the significance of the popularity of self-help texts in urban Mexico
- ☐ show how self-help texts have taken a different form in urban Mexico than in other societies
- ☐ argue that self-help texts in urban Mexico promote a collective model of family life
- ☐ explain how self-help texts have influenced the way people conduct relationships in urban Mexico

Select one answer choice.

Questions 4 and 5 are based on this passage.

Just as in many other societies, self-help texts dealing with relationships are widely sold in Mexican cities. Their widespread circulation indicates Mexicans' desire for guidance on matters of family life and other relationships in a cultural environment characterized by the erosion of traditional patriarchal cultural models. On the one hand, the widely varying accounts of and prescriptions for family life provided by these texts may be understood as reflections of the increasingly diverse and often conflicting cultural models of family life in urban Mexico. On the other hand, self-help texts may also play an important role in shaping these cultural models, insofar as they provide a substantial number of contemporary urban Mexicans with cultural options for making sense of their closest relationships.

The author mentions the "erosion of traditional patriarchal cultural models" primarily in order to

- ☐ identify an exception to the phenomenon described in the first sentence
- ☐ point out a change brought about by the phenomenon described in the first sentence
- ☐ highlight a development that facilitated the phenomenon described in the first sentence
- ☐ provide a reason why the two interpretations offered in the passage are equally likely
- ☐ mention a condition that favors one of the two interpretations offered in the passage over the other

Select one answer choice.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

Aboriginal people of Victoria, Australia, neither rejected Western material culture nor demonstrated an obsession with it; in fact, Western artifacts such as pipes and handkerchiefs were appreciated for their symbolic rather than their _____ value.

- ☐ utilitarian
- ☐ intrinsic
- ☐ communicative
- ☐ aesthetic
- ☐ pragmatic
- ☐ theoretical

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

Although genetic mutation had been fingered as the cause of the disease, recent experimental results have muddled the picture, making the link between mutation and disease seem more _____ than it once did.

- ☐ inexorable
- ☐ insidious
- ☐ enigmatic
- ☐ mysterious
- ☐ tenuous
- ☐ implacable

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

The _____ markings on some prehistoric bone and antler fragments could be accidents or the result of doodling, but may also represent a form of communication such as tally marks or even early calendars.

- ☐ purposeful
- ☐ enigmatic
- ☐ random
- ☐ inscrutable
- ☐ conspicuous
- ☐ deliberate

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

In Roth's fiction, politics is an element that doesn't naturally occur in _____ form but is always getting confused and mixed up with shame and anger, baseball and sex.

- ☐ an unalloyed
- ☐ an unbiased
- ☐ an uncontroversial
- ☐ an objective
- ☐ a neat
- ☐ a tendentious

Select two answer choices.

Questions 10 to 12 are based on this passage.

Brian Aldiss' book *Billion Year Spree* (1973), while notable for its insights, contains a weakness shared with most subsequent science fiction criticism: a failure to avoid the pitfalls of canonization. Aldiss begins properly by rejecting the total exclusion of the genre of science fiction from high literature. But he misguidedly tries to do so by adopting standards of elite criticism—that is, by establishing a canon of acceptable literature within the genre. In attempting to distinguish between real literature and “trash,” Aldiss treats popular pulp classics of the thirties, forties, and fifties condescendingly. His condescension turns to distaste when addressing their strident and (to him) hollow optimism. In place of heartiness and euphoria, he calls for the “natural and decent despair which has always characterized everyday literature.” By “everyday literature” he means, of course, mainstream writing. Because of this bias, Aldiss fails to do justice to A. E. Van Vogt, and any critical analysis of science fiction that fails to come to terms with that author's appeal and major importance defends or defines the genre by falsifying it.

The primary purpose of the passage is to

- ☐ argue that science fiction deserves something less than recognition as high literature
- ☐ challenge the appropriateness of critical analysis of science fiction as mainstream literature
- ☐ propose a refinement of an established method of analyzing themes in science fiction
- ☐ suggest an alternative approach for evaluating the literary merits of works of science fiction
- ☐ criticize a particular approach to the assessment of science fiction as literature

Select one answer choice.

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The passage suggests that Aldiss believes which of the following about popular science fiction classics of the thirties, forties, and fifties?

- ☐ They are the models on which more recent science fiction writers have based their work.
- ☐ They are characterized by a more hopeful outlook than is typical of mainstream writing.
- ☐ They constitute an important stage in the development of science fiction as a literary genre.
- ☐ They contain themes that have been largely abandoned by subsequent writers.
- ☐ They have garnered the approval of all but a small minority of critics.

Select one answer choice.

Questions 10 to 12 are based on this passage.

Brian Aldiss' book *Billion Year Spree* (1973), while notable for its insights, contains a weakness shared with most subsequent science fiction criticism: a failure to avoid the pitfalls of canonization. Aldiss begins properly by rejecting the total exclusion of the genre of science fiction from high literature. But he misguidedly tries to do so by adopting standards of elite criticism—that is, by establishing a canon of acceptable literature within the genre. In attempting to distinguish between real literature and "trash," Aldiss treats popular pulp classics of the thirties, forties, and fifties condescendingly. His condescension turns to distaste when addressing their strident and (to him) hollow optimism. In place of heartiness and euphoria, he calls for the "natural and decent despair which has always characterized everyday literature." By "everyday literature" he means, of course, mainstream writing. Because of this bias, Aldiss fails to do justice to A. E. Van Vogt, and any critical analysis of science fiction that fails to come to terms with that author's appeal and major importance defends or defines the genre by falsifying it.

The author of the passage would be most likely to agree with which of the following statements about most critics who have written about science fiction since 1973?

- ☐ They have shared a common view of the tone they expect to find in everyday literature.
- ☐ They have accorded appropriate recognition to the works of A. E. Van Vogt.
- ☐ They have repeated an error found in Aldiss' *Billion Year Spree*.
- ☐ They have been insufficiently rigorous in their treatment of works that are intended to appeal to a mass audience.
- ☐ They have been preoccupied with obscure works from the thirties, forties, and fifties.

Select one answer choice.

During a sale, the original price of a television was reduced by 25 percent. After the sale, the reduced price was reduced by 20 percent. Using a coupon, a customer purchased the television for a price that was 10 percent less than the price resulting from the two price reductions.

Quantity A

The price for which the customer purchased the television .

Quantity B

Half the original price of the television

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

$x \neq 3$ and $x \neq 5$

Quantity A

$$\frac{(x-3)(5-x)}{(3-x)(x-5)}$$

Quantity B

$$1$$

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

Select one answer choice.

Two sides of a right triangle have lengths of 5 and 10, respectively.

Quantity A

The length of the third side of the triangle

Quantity B

11

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

Select one answer choice.

A certain experiment has exactly two mutually exclusive outcomes with probabilities p and $2p$, respectively.

Quantity A

p

Quantity B

$1 - p$

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

Theo owns $\frac{1}{3}$ of the number of pairs of shoes that Samantha owns, and Richard owns twice as many pairs as Samantha. If Richard owns fewer than 30 pairs of shoes, which of the following could be the number of pairs he owns?

Indicate all such numbers.

- ☐ 29
- ☐ 24
- ☐ 18
- ☐ 15
- ☐ 12
- ☐ 10
- ☐ 9
- ☐ 6

Questions 6 to 8 are based on the following data.

Hours Billed by Employees of Company X, 1996–2000**Distribution of Hours Billed by Employee Job Category, 2000**

For the employees in the customer service job category in 2000, the average billing rate was \$18.50 per hour. Which of the following is the best estimate of the total amount billed by employees in the customer service job category in 2000?

☐ \$12,000,000

☐ \$1,200,000

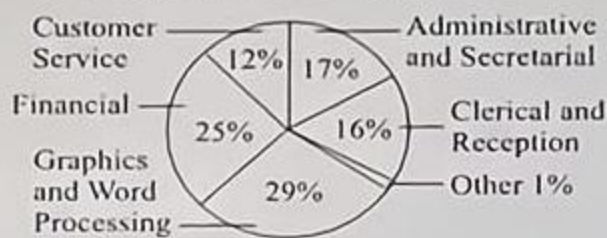
☐ \$120,000

☐ \$12,000

☐ \$1,200

Select one answer choice.

Questions 6 to 8 are based on the following data.

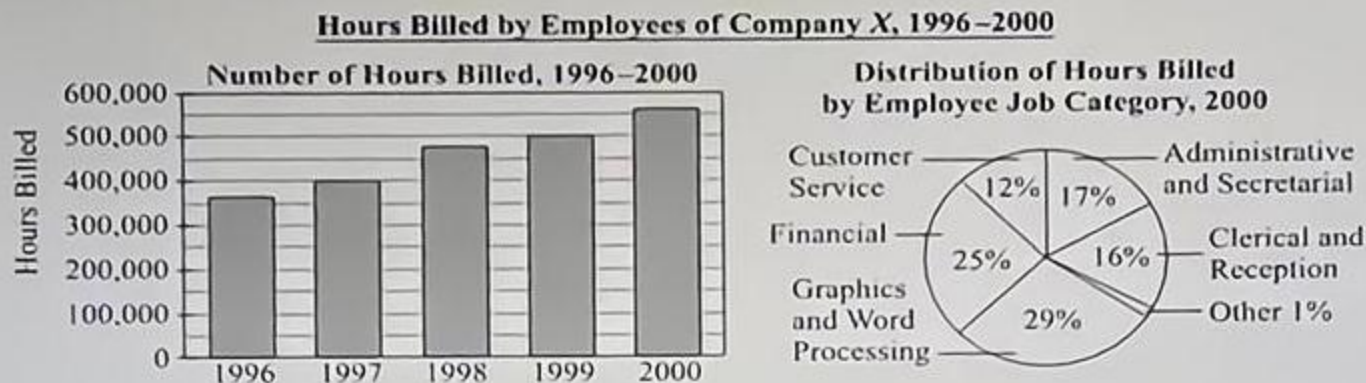
Hours Billed by Employees of Company X, 1996–2000**Distribution of Hours Billed
by Employee Job Category, 2000**

Which of the following is closest to the ratio of the number of hours billed by employees in 1999 to the number of hours billed by employees in 1996 ?

☐ 2 to 3☐ 3 to 4☐ 4 to 3☐ 5 to 3☐ 2 to 1

Select one answer choice.

Questions 6 to 8 are based on the following data.



If the number of hours billed by employees in the customer service job category was the same in 1999 as it was in 2000, which of the following is closest to the percent of the total number of hours billed by employees in 1999 that was billed by employees in the customer service job category in 1999?

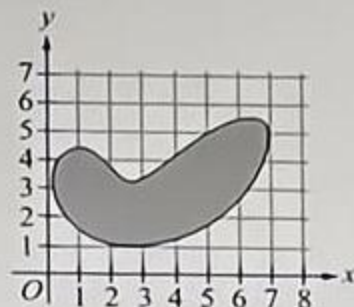
- ☐ 19% ☐ 17% ☐ 15% ☐ 13% ☐ 12%

Select one answer choice.

If n and p are positive integers and p is a divisor of n , how many positive integers less than or equal to n are divisible by p ?

- ☐ 1
- ☐ p
- ☐ n
- ☐ $\frac{n}{p}$
- ☐ np

Select one answer choice.



The shaded region in the xy -plane represents the surface of a pond. A length of 1 unit in the xy -plane represents 5 meters. The average depth of the water in the pond is 0.8 meter. Of the following, which is the best estimate of the total volume, in cubic meters, of the water in the pond?

- ☐ 100 ☐ 350 ☐ 750 ☐ 1,000 ☐ 1,300

Select one answer choice.

If $y = 2x + 3$, then which of the following intervals consists of the values of x such that $xy < 0$?

- ☐ $-\frac{3}{2} < x < 0$ ☐ $-\frac{3}{2} < x < \frac{3}{2}$ ☐ $-3 < x < 2$ ☐ $0 < x < \frac{3}{2}$ ☐ $2 < x < 3$

Select one answer choice.

The width of rectangle R is $\frac{3}{7}$ of the length. If the area of R is 84, what is the perimeter of R ?

Enter your answer as an integer or a decimal in the answer box. Backspace to erase.

Select one entry for the blank. Fill the blank in the way that best completes the text.

The results of the latest survey being _____, earlier results gave researchers little relevant experience for the new data' s interpretation.

superfluous
inconsequential
tedious
unexampled
familiar

Select one answer choice.

The moral sentiments expressed in Aesop' s fables are not considered to be imaginative today, and in the seventeenth century they were similarly regarded as _____.

pedestrian
didactic
grandiose
germane
rhetorical

For each blank select one entry from the corresponding column of choices. Fill all blanks in the way that best completes the text.

At one time researchers viewed the placebo effect as (i)_____, a statistical (ii)_____ faced by those attempting an objective evaluation of the efficacy of potentially legitimate therapies. That view has changed: the placebo effect is today seen as an important part of the healing process.

Blank (i)

a boon
a touchstone
an impediment

Blank (ii)

increment
annoyance
deviation

Select one entry from each column.

For each blank select one entry from the corresponding column of choices. Fill all blanks in the way that best completes the text.

A recent survey of 120 United States corporations found that a third of employees in blue-chip corporations wrote poorly in their electronic mail. Millions of these (i) _____ e-mail messages are clogging computers by setting off requests for (ii) _____, and many of the requests, in turn, are also (iii) _____ written, resulting in whole cycles of confusion.

Blank (i)

litigious
inscrutable
intuitive

Blank (ii)

evaluation
elucidation
authentication

Blank (iii)

decisively
shoddily
unresponsively

Select one entry from each column.

Questions 5 and 6 are based on this passage.

Margaret Bayard Smith's 1836 biographical sketch of Dolley Madison extensively quotes a letter Madison allegedly wrote to her sister in 1814 detailing her last hours in the White House before it was burned by invading troops. Madison, who gave Smith a copy of the letter, provided conflicting accounts of the original document's whereabouts. It is possible that in 1814 Madison wrote the letter Smith quoted, or one very similar: other authenticated documents she later wrote agree with its account. More likely, Madison may have edited or embellished the original letter for publication. In contrast to her typically slapdash style, the letter given to Smith unfolds like a novel, in uncharacteristically formal and flowery language. The text also includes details unnecessary in a letter between sisters, though useful to general readers.

The author of the passage mentions "other authenticated documents" primarily in order to

- ☐ corroborate Smith's account of how she came into possession of a copy of Madison's letter
- ☐ point out the contrast between the verifiability of the copy of the letter Madison gave Smith and other documents Madison wrote
- ☐ show that Madison did not usually give conflicting accounts of the whereabouts of documents she had produced
- ☐ suggest that the account in the copy of the letter Madison gave to Smith is consistent with the account in the original letter
- ☐ provide evidence that supports the passage's characterization of Madison's usual writing style

Select one answer choice.

Questions 5 and 6 are based on this passage.

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Consider each of the choices separately and select all that apply.

It can be inferred that the author of the passage finds which of the following factors relevant in evaluating the letter Madison gave to Smith?

- ☐ Inconsistencies in Madison's statements about the original letter
- ☐ The writing style of the letter that Madison gave to Smith
- ☐ The apparent intended audience of the letter that Madison gave to Smith

Select one or more answer choices.

Questions 7 to 9 are based on this passage.

For the past twenty years, most interpretations of Native American literature have focused on either theories, texts, or territories. Theories of reading, such as postcolonialism, postmodernism, the oral tradition, and cultural recovery, have served as major modes of inquiry for both Native American and non-Native American critics who try to make sense of Native American literary texts. Other scholars put the texts themselves in the foreground. In the case of Native American literature, text-based approaches accentuate the work of writers such as Silko, Momaday, and Ortiz over a particular theory or idea. A third group of scholars root their analysis of texts, whether they be myths, songs, poems, or prose, in the land that is fundamental to the works under consideration. Rather than focusing on a theory or a writer, a critic engaged in a territory-based reading focuses on the healing or recuperative or symbolic properties of culturally specific geographical locations. None of these strategies is any more important than another, and it is impossible to conceive of Native American literary criticism without each, as all three contribute to understanding the complexity of Native American discourses.

The primary purpose of the passage is to

- ☐ propose a new way of understanding a particular literary tradition
- ☐ advocate a synthesis of various approaches to the study of a particular literary tradition
- ☐ identify leading strategies used to interpret a particular literary tradition
- ☐ discuss important methodological similarities among various interpretive approaches to a particular literary tradition
- ☐ distinguish between opposing critical approaches to a particular literary tradition

Select one answer choice.

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The passage suggests that a theory-based approach to Native American literature and a territory-based approach have which of the following characteristics in common?

- ☐ Neither is exclusively concerned with the text itself.
- ☐ Neither is well suited to writers such as Silko, Momaday, or Ortiz.
- ☐ Both focus primarily on the act of reading.
- ☐ Both are strategies that work better with some literary genres than with others.
- ☐ Both have recently gained in popularity at the expense of text-based approaches.

Select one answer choice.

Questions 7 to 9 are based on this passage.

For the past twenty years, most interpretations of Native American literature have focused on either theories, texts, or territories. Theories of reading, such as postcolonialism, postmodernism, the oral tradition, and cultural recovery, have served as major modes of inquiry for both Native American and non-Native American critics who try to make sense of Native American literary texts. Other scholars put the texts themselves in the foreground. In the case of Native American literature, text-based approaches accentuate the work of writers such as Silko, Momaday, and Ortiz over a particular theory or idea. A third group of scholars root their analysis of texts, whether they be myths, songs, poems, or prose, in the land that is fundamental to the works under consideration. Rather than focusing on a theory or a writer, a critic engaged in a territory-based reading focuses on the healing or recuperative or symbolic properties of culturally specific geographical locations. None of these strategies is any more important than another, and it is impossible to conceive of Native American literary criticism without each, as all three contribute to understanding the complexity of Native American discourses.

The author of the passage mentions postcolonialism, postmodernism, the oral tradition, and cultural recovery primarily in order to

- ☐ identify certain theories of reading that the author believes are more sophisticated than others
- ☐ identify certain interpretive strategies that are most effective when used in conjunction with other strategies
- ☐ identify the modes of inquiry that have been used most successfully by both Native American and non-Native American critics of Native American literature
- ☐ provide examples of theoretical approaches that are incompatible with more text-based approaches
- ☐ provide examples of one of three major types of interpretive strategies that are applied to Native American literature

Select one answer choice.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

In her career as an editor, she pruned and shaped many a writer' s _____ prose into crisp lucidity.

- ☐ wayward
- ☐ transparent
- ☐ errant
- ☐ urbane
- ☐ elegant
- ☐ incisive

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

It is true that science, and more particularly scientists, _____ cherished paradigms with great reluctance and that when they do, scientific revolutions may result.

- ☐ unify
- ☐ share
- ☐ cede
- ☐ embrace
- ☐ dismantle
- ☐ relinquish

Select two answer choices.

Select the two answer choices that, when used to complete the sentence, fit the meaning of the sentence as a whole and produce completed sentences that are alike in meaning.

When urged to give up his place in line, the child initially _____ but eventually acquiesced.

- ☐ retaliated
- ☐ capitulated
- ☐ commiserated
- ☐ remonstrated
- ☐ complied
- ☐ protested

Select two answer choices.

Question 13 is based on this passage.

Belnovia's government provides health care to all its citizens. Five years ago, with costs incurred from smoking-related health problems accounting for 5 percent of annual government expenditures, Belnovia's health agency initiated a nationwide antismoking campaign. Although the campaign succeeded in reducing the number of smokers in Belnovia substantially, it clearly failed to reduce the annual costs incurred from smoking-related health problems, since these costs currently account for 6 percent of annual government expenditures.

Which of the following is an assumption on which the argument depends?

- ☐ Costs incurred from smoking-related health problems account for a greater portion of the Belnovian government's overall health expenditures now than they did five years ago.
- ☐ The average cost of treatment for smoking-related health problems in Belnovia has not increased significantly over the past five years.
- ☐ Overall health expenditures account for a greater portion of the Belnovian government's annual expenditures now than they did five years ago.
- ☐ Belnovia's annual government expenditures are not significantly lower now than they were five years ago.
- ☐ Smoking-related health problems usually persist long after a person has quit smoking.

Select one answer choice.

Questions 14 and 15 are based on this passage.

Invasions of nonnative species are clearly reducing species diversity on a global scale. On the regional scale, these invasions cause a change in species composition, but the impact of these invasions on diversity is less certain. Changes in diversity within a region depend on the number of extinctions of native species. In some cases, broad losses of native diversity have followed the establishment of nonnative species; for example, the introduction of the brown tree snake led to the extinction of several bird species on Guam. In other cases, many nonnative species have had no detectable effects on native flora and fauna. Consequently, in regions where many nonnative species have become established without coincident extinctions, diversity has increased.

Which of the following best describes the function of the highlighted sentence within the context of the passage as a whole?

- ☐ It calls into question a distinction made in the previous sentence.
- ☐ It responds to a criticism made in the previous sentence.
- ☐ It helps explain an assertion made in the previous sentence.
- ☐ It introduces a problem that is discussed later in the passage.
- ☐ It anticipates a dispute that is discussed later in the passage.

Select one answer choice.

Questions 14 and 15 are based on this passage.

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Consider each of the choices separately and select all that apply.

The author of the passage would be likely to agree with which of the following statements about the effects of invasions by nonnative species on species diversity?

- ☐ Invasions of nonnative species have led to an overall reduction in global species diversity.
- ☐ In regions where several native species have become extinct following the invasion of a nonnative species, the diversity of those regions has decreased.
- ☐ In regions where invasions of nonnative species have not led to extinctions of native species, the diversity of those regions has increased.

Select one or more answer choices.

	Number of Scores	Arithmetic Mean	Standard Deviation
Class I	35	72	8
Class II	42	72	10

Students in two different classes took the same English test. The table shows statistical results calculated on the students' scores.

Quantity A

The median score for Class I

Quantity B

The median score for Class II

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

Select one answer choice.

Line ℓ in the xy -plane contains the points $(a, 0)$ and $(0, b)$, where $b < a < 0$.

Quantity A

The slope of line ℓ

Quantity B

-1

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

Select one answer choice.

The area of a circle is π^5 .

Quantity A

The circumference of the circle

Quantity B

$$2\pi^3$$

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

Select one answer choice.

$$\frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, -\frac{1}{16}, \dots$$

In the infinite sequence shown, each term after the first is found by multiplying the previous term by $-\frac{1}{2}$.

Quantity A

The 56th term

Quantity B

The 60th term

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

In the rectangular coordinate system, the point $(0, 0)$ is equidistant from the points $(x, 0)$ and $(0, y)$.

Quantity A

x

Quantity B

y

- ☐ Quantity A is greater.
- ☐ Quantity B is greater.
- ☐ The two quantities are equal.
- ☐ The relationship cannot be determined from the information given.

Select one answer choice.



When the positive integer y is divided by the positive integer x , the quotient is 7 and the remainder is 2. What is y in terms of x ?

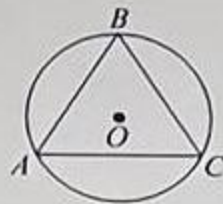
- ☐ $7(x+2)$ ☐ $7x+2$ ☐ $7+\frac{2}{x}$ ☐ $7x-2$ ☐ $7(x-2)$

Select one answer choice.

For a list of k consecutive integers, the median is m and the range is r . Which of the following must be equal to k ?

- ☐ m
- ☐ r
- ☐ $m + 1$
- ☐ $r + 1$
- ☐ $r - m + 1$

Select one answer choice.



Equilateral triangle ABC is inscribed in a circle with center O and area 64π . What is the height of triangle ABC ?

- ☐ 8
- ☐ 10
- ☐ 12
- ☐ 14
- ☐ 16

Select one answer choice.

Stations A , M , and B are located along a certain train route, and Station M is between Stations A and B . At noon, a train engine passed Station A traveling at a constant speed of 80 kilometers per hour toward Station B . Also at noon, another train engine passed Station B traveling at a constant speed of 60 kilometers per hour toward Station A . Both train engines passed Station M at the same time. What is the ratio of the distance along the route between Stations A and M to the distance along the route between Stations A and B ?

☐ $\frac{1}{4}$

☐ $\frac{3}{7}$

☐ $\frac{1}{2}$

☐ $\frac{4}{7}$

☐ $\frac{3}{4}$

Select one answer choice.

Which of the following is equal to $(6)(14^8) + (15)(14^7)$?

☐ $(21)(14^7)$

☐ $(35)(14^7)$

☐ $(84)(14^7)$

☐ $(90)(14^7)$

☐ $(99)(14^7)$

Select one answer choice.

A pound of tea makes 210 cups of tea, and a pound of coffee makes 40 cups of coffee. If, on the average, a restaurant serves 12 times as many cups of coffee as tea, what is the ratio of pounds of tea to pounds of coffee used at the restaurant?

☐ $\frac{1}{63}$

☐ $\frac{6}{125}$

☐ $\frac{1}{12}$

☐ $\frac{4}{21}$

☐ $\frac{16}{7}$

Select one answer choice.

What are the possible values of $x^2 + 7x + 8$ if x satisfies the equation $x^2 + 3x - 4 = 0$?

Indicate all such values.

☐ -6

☐ -4

☐ 2

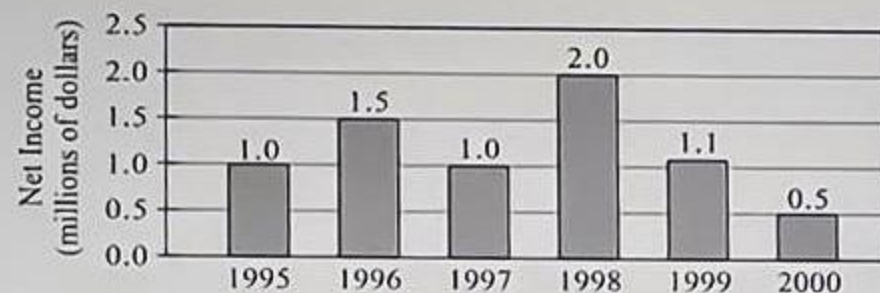
☐ 8

☐ 16

☐ 38

☐ 52

Select one or more answer choices.



The graph shows the net income of a company for each of 6 years. For which of the following one-year time intervals was the percent increase or percent decrease in the net income greater than or equal to 50 percent?

Indicate all such one-year time intervals.

- ☐ 1995–1996
- ☐ 1996–1997
- ☐ 1997–1998
- ☐ 1998–1999
- ☐ 1999–2000

Select one or more answer choices.

In a group of 10 people, exactly 4 are vegetarians. A committee of 3 will be formed by randomly selecting 3 people from the group. How many of the possible 3-member committees would include exactly 2 vegetarian members?

- ☐ 3
- ☐ 6
- ☐ 15
- ☐ 24
- ☐ 36

Select one answer choice.

A deposit of \$3,000 was made into an account that earned interest at the annual rate of r percent, compounded annually, where r is a constant. The interest was paid into the account, and there were no other deposits to or withdrawals from the account. At the end of the first year, the amount in the account, including interest, was \$3,090. The amount of interest earned for the second year was what fraction of the total amount of interest earned for the 2 years combined?

Enter your answer as a fraction, with the numerator and denominator in their respective answer boxes. Backspace to erase.

section 2	1	2	3	4	5	6	7	8	9	10	11	12			
答案	A	CA	ABC	B	C	AE	CD	BD	AE	E	B	C			
section 3	1	2	3	4	5	6	7	8	9	10	11	12			
答案	A	C	D	B	24,18,12,6	1,200,000	C	D	D	B	A	40			
section 4	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
答案	D	A	CB	BBB	A	BC	C	A	E	AC	CF	DF	D	C	ABC
section 5	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
答案	D	B	C	B	D	B	D	C	D	E	A	16,-4	ACE	E	103/203